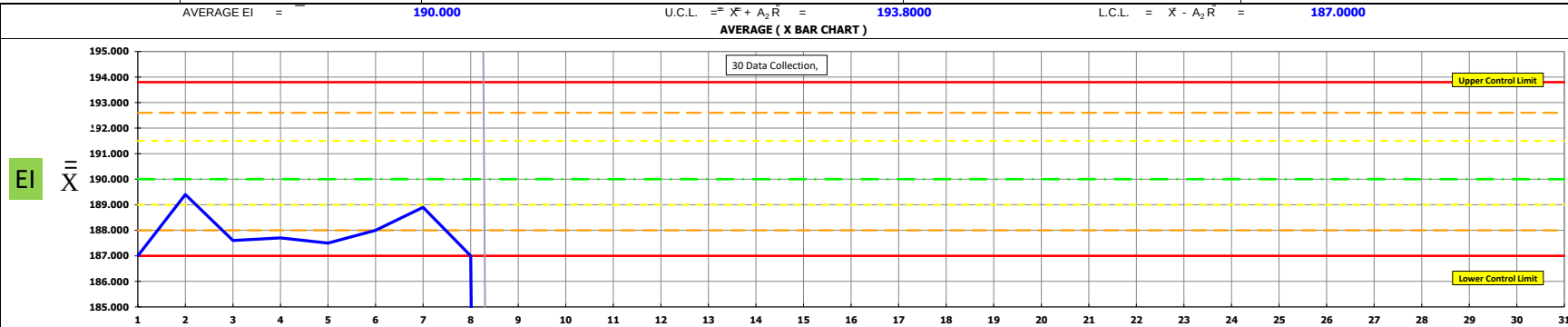


$\bar{X}$ - R CONTROL CHART

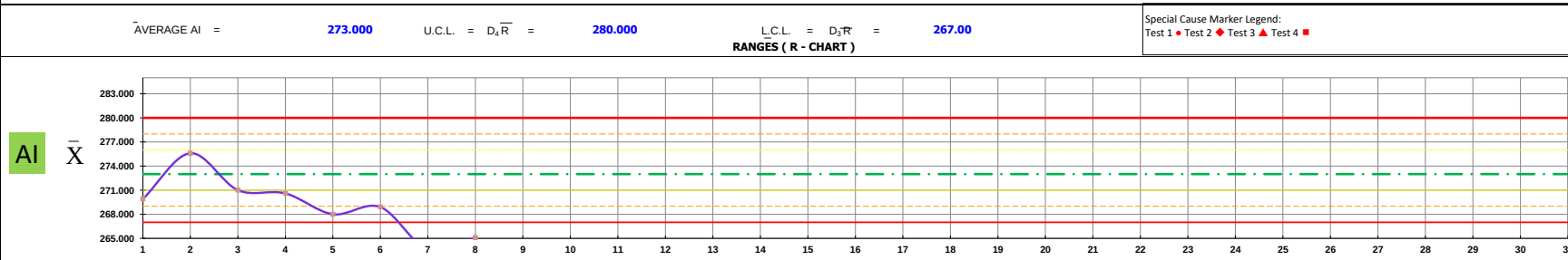
No. Dokumen	4-QAS-PP07-001
Revisi	00
Tgl Berlaku	2-Dec-16

MACHINE M-Tan	SECTION QC	OPERATION M-Tan	DATE CONTROL LIMITS CALC.	SPECIFICATION min EI - : 180 Al Min : 260	COURSE NO.	CONTROL ITEM (Δ) YES NO	☐
	MONTH Jul-25	CHARACTERISTIC Temperatur		SAMPLE SIZE / FREQUENCY n = 3	TYPE / SIZE : D21DW4-B-2 / 0.70		

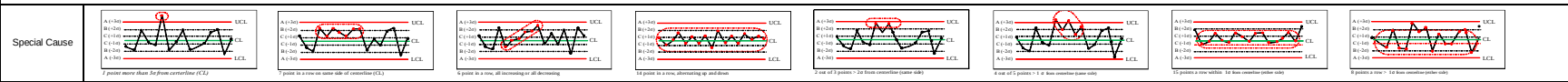


- | SPECIAL CAUSES | |
|----------------|--|
| ✓ | TEST 1: ANY POINT OUT OF CONTROL LIMIT |
| ✓ | TEST 2: 7 POINTS IN A ROW ON SAME SIDE OF CENTER LINE |
| ✓ | TEST 3: 6 POINTS IN A ROW ALL INCREASING OR DECREASING |
| ✓ | TEST 4: 14 POINTS IN A ROW, ALTERNATING UP & DOWN |
| ✓ | TEST 5: 2 OUT OF 3 POINTS > 2 STDEV FROM CENTER LINE (SAME SIDE) |
| ✓ | TEST 6: 4 OUT OF 5 POINTS > 1 STDEV FROM CENTER LINE (SAME SIDE) |
| ✓ | TEST 7: 15 POINTS IN A ROW WITHIN 1 STDEV OF CENTER LINE (EITHER SIDE) |
| ✓ | TEST 8: 8 POINTS IN A ROW > 1 STDEV FROM CENTER LINE (EITHER SIDE) |

ACTION INSTRUCTION	
1.	_____
2.	_____
3.	_____
4.	_____
5.	_____



SUBGROUP SIZE	A_2	d_2	D_2	D_4
2	1.881	1.128	0	3.267
3	1.023	1.693	0	2.574
4	0.729	2.059	0	2.281
5	0.577	2.326	0	2.115
6	0.483	2.534	0	2.004
7	0.4193	2.704	0.0756	1.9244
8	0.3726	2.847	0.1361	1.8639

[illegible]

CALCULATION	
$EI(X) = 187.9$ $AI(X) = 269.1$ $AI(StdDev) = 3.7 \quad StdDev = 0.859$	
THE PROCESS MUST BE IN CONTROL BEFORE CAPABILITY CAN BE DETERMINED	
$C_{PK} = \min \left[\frac{X - LSL}{3\sigma_{EST}}, \frac{USL - X}{3\sigma_{EST}} \right]$ $= \min \left[\frac{0.34}{2.29}, \frac{1.0}{1.0} \right]$ $= 0.34$	1.0 1.0
$C_p = \left[\frac{USL - LSL}{6\sigma_{EST}} \right]$ $= \left[\frac{EI}{1.32} \right]$	AI 0.58
CHECKED BY	APPROVED BY